Proxy Features and Fairness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach distinct claims with distinct evidence.

Supply every field origin and paired prediction,keep retained proxies visible and avoid inferring real identities or unobserved consequences.

Keep one habit: Require contextual evaluation goals,source counts and unresolved risks beside any mitigation statement.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Identify possible **proxy feature**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach two 45-minute sessions. This week focuses on **proxy feature**s and **fairness**. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

No; examine errors, consequences, data and context.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

Opening retrieval

Without opening the old lesson, explain one key idea from Subgroup Performance. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate subgroup results with sample sizes

Reference example, not the only acceptable wording: Teacher explanation: A 2/20=10 percent; B 3/5=60 percent; pooled 5/25=20 percent, a 50-point group gap.

Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

Look for: Create features available at **prediction** time

Reference example, not the only acceptable wording: Teacher explanation: Areas 6, 8, 6 square cm; **training** mean 20/3 square cm.

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Record enough detail to reproduce a run

Reference example, not the only acceptable wording: Teacher explanation: Mean 70 percent; range 80-60=20 percentage points, across the three declared runs.

End goal

By the end, I can explain **proxy feature**s and **fairness** using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Subgroup Performance

Feature Engineering

Reproducible Experiments

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A leaf model performs well using location ID alone. Explain a possible proxy problem and design a diagnostic evaluation.

Concept to explain

A feature may indirectly encode sensitive identity or context, creating unjustified differences even when the original field is absent.

Term	Meaning
proxy feature	input correlated with another sensitive attribute
fairness	evaluate benefits and harms across affected groups
paired test	cases differing in one relevant factor
mitigation	control reducing an identified risk

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach proxy association separately from actual model effects

> Keep the imperfect main association,perfect pictured mapping and executable ablation distinct.

Main scope and complete association table

An **input** can carry information about a characteristic without naming it. In a fictional school-device dataset, device code may be associated with building or access to newer equipment.

Removing a building field does not remove all related information. Do not infer a real student identity or protected characteristic in this exercise.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **proxy feature**s and **fairness**.

Complete fictional association table	DeviceX	Other device	Total
North building	8	2	10
South building	2	8	10
Total	10	10	20

Fictional records show device code X on 8 of 10 north-building examples and 2 of 10 south-building examples. Device and building are associated in this sample.

This does not prove device causes model errors or that every X device comes from north. The remaining cases contradict such a universal rule.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Term	Meaning
Proxy feature	A variable that can stand in for another characteristic
Correlation	An observed association, not proof of cause
Sensitive attribute	A characteristic requiring particular care in a context
Ablation	A comparison that removes a feature or component

Proxy features and fairness at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
North examples	8 X of 10	2 other devices
South examples	2 X of 10	8 other devices
Review question	What does code represent?	Check context and errors

A feature may be legitimate for one purpose and inappropriate for another. Ask what is measured, when it is available, why it predicts and who might be disadvantaged. A correlation alone neither authorizes use nor proves harm. Talk through the intended decision and available alternatives.

Why might removing building but keeping device code leave related information? Explain the steps or evidence behind your response.

Read separate pictured retained-information mapping

Removing an obvious setting field may leave another field that carries the same information.

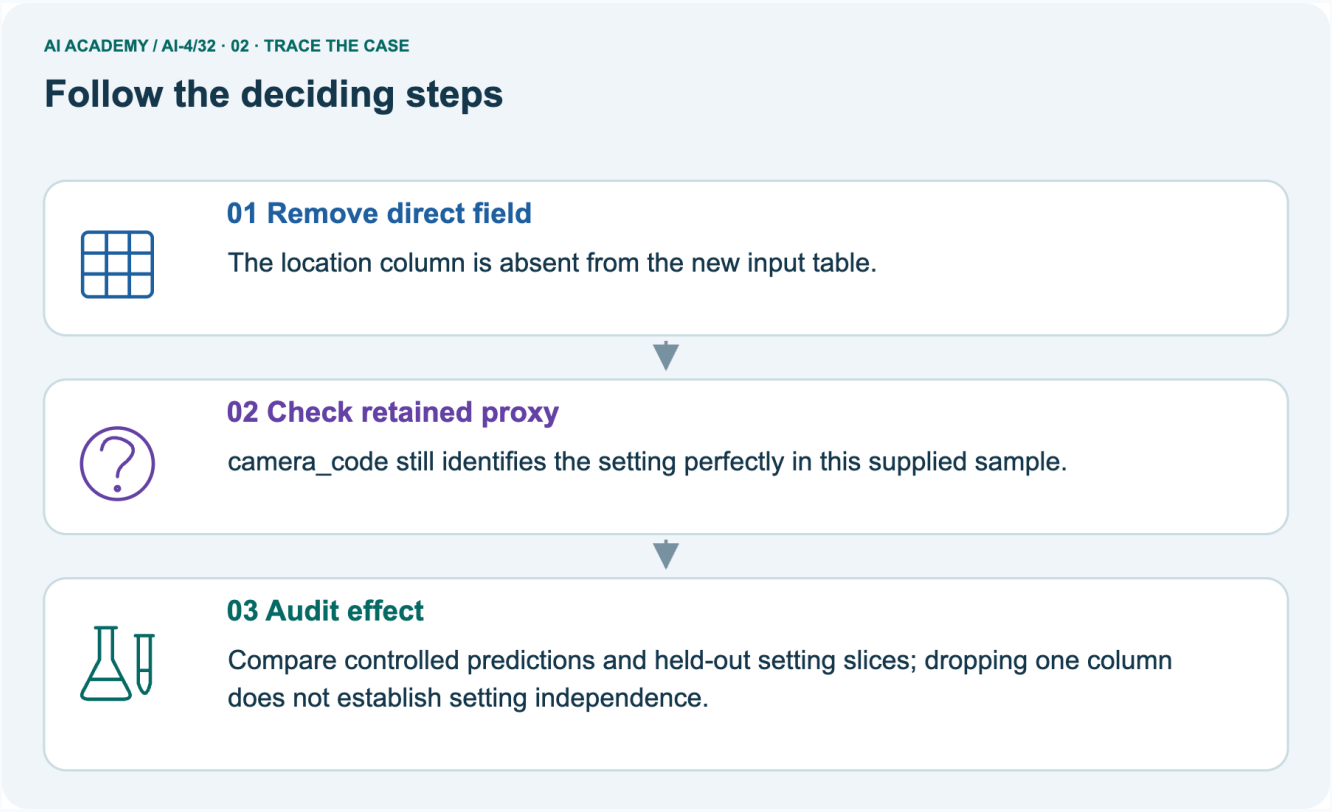
Toy table has location=greenhouse/desk and camera_code=G/D. Every G record is greenhouse; every D record is desk. The team drops location.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture’s original toy mapping is perfect:cameraGidentifies greenhouse and cameraDdesk in that supplied dataset, even after location is dropped. The main North/Southassociation is only 8/10 versus2/10, not perfect. A new balanced crossed camera/setting dataset breaks the old one-to-one mapping, but that does not prove every other retained field is independent of setting.

The complete ablation trial below intentionally retains room_hintas another setting proxy; its constructed marker/label relation is not a real diagnostic or fairness result.

Trace deletion,crossed assignment and remaining-risk limits



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Remove direct field	The location column is absent from the new input table.
2. Check retained proxy	camera_code still identifies the setting perfectly in this supplied sample.
3. Audit effect	Compare controlled predictions and held-out setting slices; dropping one column does not establish setting independence.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

A correlated field is not necessarily harmful in every use; assess task relevance, consequences and actual model behaviour.

Change: Camera codes are assigned independently of setting in a new balanced dataset.

Prediction: camera_code no longer perfectly identifies setting in that dataset.

Reason: The observed mapping changed; other retained fields may still contain setting information.

▼ Additional background/location risk scenario

Background colour tracks collection location, associated with plant types.

Explain proxy risk and test.

Background can be a location shortcut rather than leaf **evidence**. Test leaf types across backgrounds and locations and compare slices.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Trace suspicious proxies and compare model behavior after a controlled removal.

MODEL IDEA A feature may indirectly encode sensitive identity or context, creating unjustified differences even when the original field is absent.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Photo background predicts classroom location and is correlated with camera quality. Model uses background as a shortcut for health labels.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Photo background predicts classroom location and is correlated with camera quality.

STEP 2 Model uses background as a shortcut for health labels.

STEP 3 Create paired crops with the same leaf and different backgrounds; **prediction**s change.

STEP 4 Crop backgrounds or improve data, retrain, and compare performance and subgroup gaps.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Exact explanation to model

Explain the added worked example

Key terms

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

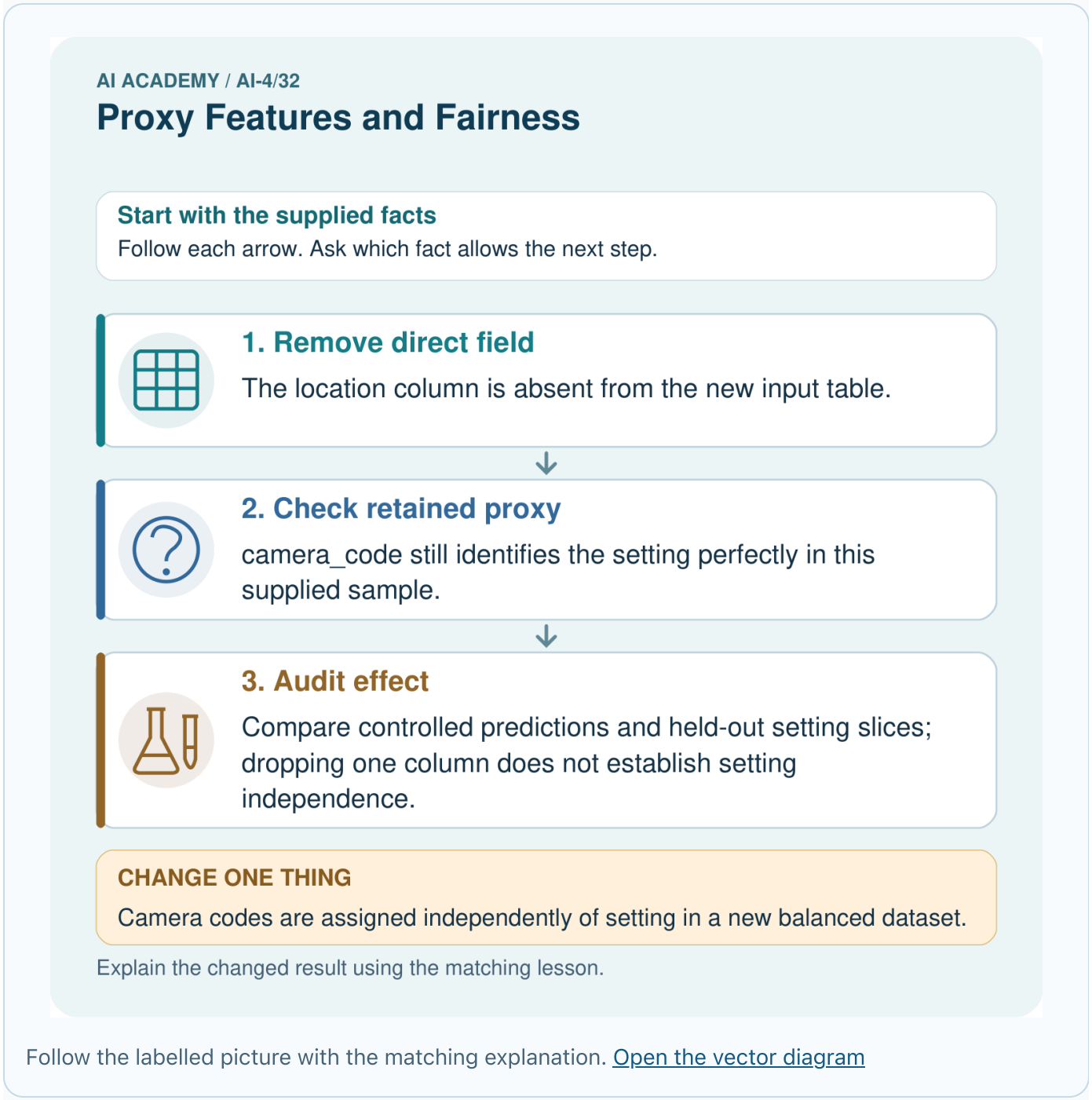
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Proxy Features and Fairness

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: Why might removing building but keeping device code leave related information?

Teacher check: Teacher explanation: Device code is associated with building in the sample and may carry some of that information indirectly.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Proxy feature**s and **fairness**. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: X occurs 8/10 north and 2/10 south in this sample.

This supports an association, not a deterministic identity rule or a causal explanation.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

What **evidence** contradicts the original “all setting information removed” claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

The retained G/D code maps exactly to greenhouse/desk, so the supplied table still exposes setting information. Changed-case check: camera_code no longer perfectly identifies setting in that dataset.

The observed mapping changed; other retained fields may still contain setting information.

Changed-case reasoning: camera_code no longer perfectly identifies setting in that dataset. The observed mapping changed; other retained fields may still contain setting information.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: Background colour tracks collection location, associated with plant types.

Write the question to explain: Explain proxy risk and test.

Reveal after **prediction**s: Background can be a location shortcut rather than leaf **evidence**. Test leaf types across backgrounds and locations and compare slices.

Use the earlier transfer question as additional practice. Reserve the selected fresh postcode/school-code evidence-request task for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided response: Device code is associated with building in the sample and may carry some of that information indirectly. Exit response: No.

Other features can retain information and accuracy can hide different errors and consequences.

Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: Does equal overall accuracy prove fair treatment? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify proxy feature in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: read an association; find a proxy risk. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between proxy feature and correlation. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Describe a controlled with/without-code comparison.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Identify possible proxy features; Explain why removing one field does not guarantee fairness. Record any support used.

Stage: 35 to 45 minutes

> **Teacher action:** Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Ablation fixture and limits as supplied. Removing camera leaves room_hint, so same 4/8 accuracy can hide changed recall and retained context. Paired P1 greenhousecamera1/0 showsV1 sensitivity while V2 usesroom. Main8/10 versus2/10 association is imperfect; do not invent identities or harms.

Independent postcode-removal claim unsupported because school code may relate to location; request provenance, actual associations, context, matched version/group errors and unresolved proxy checks. No observed mapping or fairness result supplied.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. Proxy Detective: map feature origins, run **paired test**s, remove one suspicious feature, and check both utility and gaps.
3. 3. Run one normal, one boundary, one missing, and one shifted case.

4. 4. Calculate the metric from raw counts or row-level errors.

5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: X occurs 8/10 north and 2/10 south in this sample.

This supports an association, not a deterministic identity rule or a causal explanation.

A stronger claim makes a stronger project: No. Other features can retain information and accuracy can hide different errors and consequences.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: The claim is unsupported: school code may be associated with location.

Review provenance and associations, compare **ablation**s and group errors, and state unresolved proxy risks.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the selected fresh postcode/school-code evidence-request task for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR Removing a sensitive column guarantees the model cannot use related information.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: A team removes postcode but retains school code and claims all location-related information is gone. Evaluate the claim and request **evidence**.

Teacher explanation: The claim is unsupported: school code may be associated with location. Review provenance and associations, compare **ablation**s and group errors, and state unresolved proxy risks.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Read an association

What do the device counts support, and what do they not support?

Teacher explanation: X occurs 8/10 north and 2/10 south in this sample. This supports an association, not a deterministic identity rule or a causal explanation.

W2 Find a proxy risk

Why might removing building but keeping device code leave related information?

Teacher explanation: Device code is associated with building in the sample and may carry some of that information indirectly.

W3 Plan an ablation

Describe a controlled with/without-code comparison.

Teacher explanation: Use the same validation rows and metric, fit both versions only on **training**, and report overall and relevant subgroup errors with denominators.

W5 Limit the result

Does unchanged accuracy after removing a proxy prove **fairness**?

Teacher explanation: No. Other features can retain information and accuracy can hide different errors and consequences.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Assess and explain the answers

Worked case answer

Background can be a location shortcut rather than leaf **evidence**. Test leaf types across backgrounds and locations and compare slices.

Additional transfer practice question

A plant classifier uses camera_model, and one species was photographed mostly on a single camera. What shortcut might it learn and how could you investigate?

Reasoned practice answer

It may use camera artifacts as a proxy for species. Test across cameras with balanced species coverage and compare a model excluding the feature.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
</details>

Criterion	0 points	1 point	2 points
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A feature unrelated to plant biology cannot carry label-correlated information.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Location may encode class prevalence rather than leaf **evidence**. Evaluate leaf classes across new and shared locations, inspect coverage and use appropriate grouping.

A correlated proxy can fail after deployment; high existing accuracy does not prove a useful causal feature.

Explanations for the additional workbook examples

W1 Read an association

W2 Find a proxy risk

W3 Plan an ablation

W4 Apply independently

A team removes postcode but retains school code and claims all location-related information is gone. Evaluate the claim and request **evidence**.

Teacher explanation: The claim is unsupported: school code may be associated with location. Review provenance and associations, compare **ablation**s and group errors, and state unresolved proxy risks.

W5 Limit the result

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Document Plant Classifier proxy risks, paired-test **evidence**, **mitigation**, and remaining **fairness** limitation.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **proxy feature** and **correlation** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: The claim is unsupported: school code may be associated with location. Review provenance and associations, compare **ablation**s and group errors, and state unresolved proxy risks.

Fresh retry: removing a device-brand field while retaining model number may leave brand-related information. Require a contextual association check and no automatic **fairness** claim.

Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Delayed return

Suggested return points in this level: ai-4/33, ai-4/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 33: Model Cards and Human Review. Retrieve today's **evidence** and identify what carries forward and what is new.

Next week: Model cards and review. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case
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<details>
<summary>Primary assessment-context reference</summary>

Teacher context reference: Fairlearn assessment guidance starts from possible harms and relevant groups before selecting/disaggregating metrics: https://fairlearn.org/main/user_guide/assessment/index.html . This lesson uses fictional object/setting records, not actual personal attributes or a complete fairness assessment.
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Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Identify possible **proxy feature**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Why might removing building but keeping device code leave related information?

Reasoned practice answer: Teacher explanation: Device code is associated with building in the sample and may carry some of that information indirectly.

Check every learner before discussion: Use the worked case for **Proxy feature**s and **fairness**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: X occurs 8/10 north and 2/10 south in this sample.

This supports an association, not a deterministic identity rule or a causal explanation.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Camera codes are assigned independently of setting in a new balanced dataset. Require a **prediction** and a reason before comparing. Expected result: camera_code no longer perfectly identifies setting in that dataset.

Reason: The observed mapping changed; other retained fields may still contain setting information.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

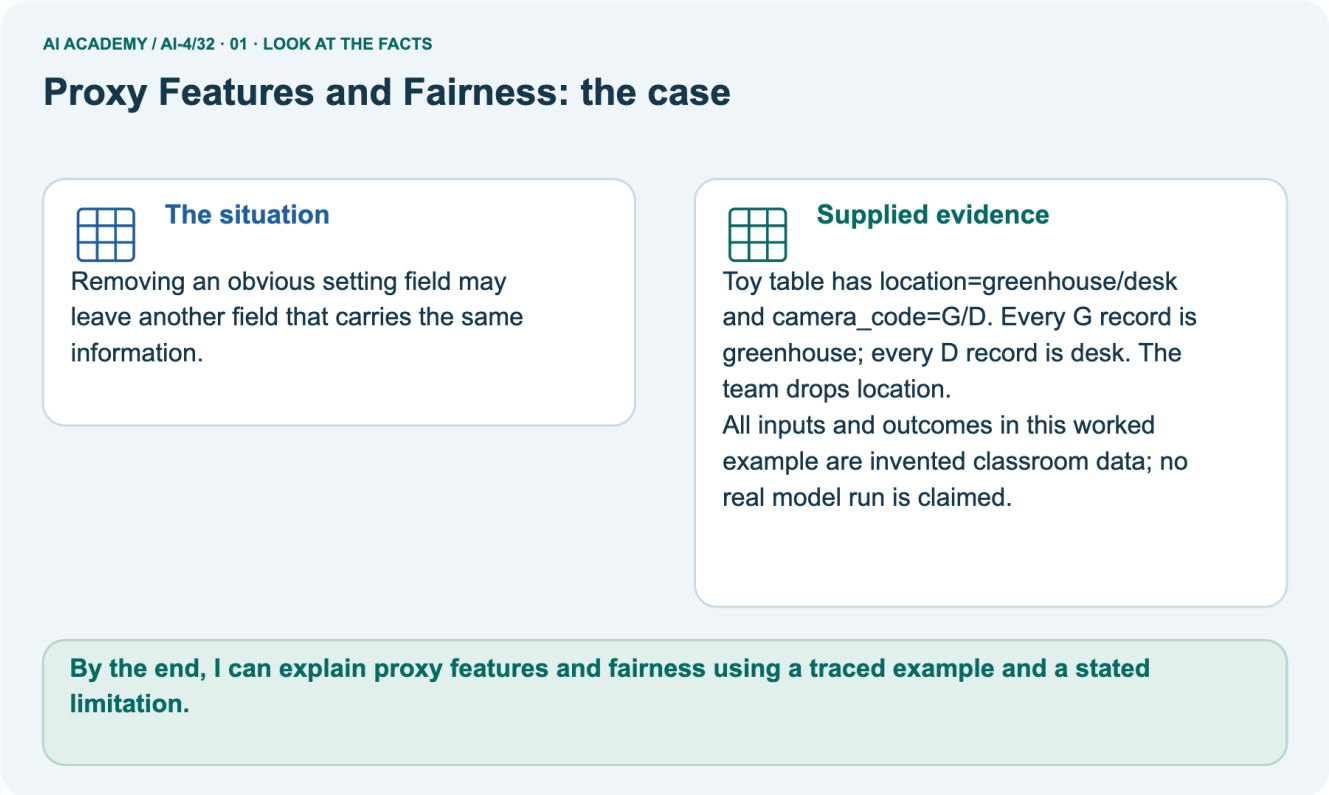
<details>
<summary>Reference · Teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Proxy Features and Fairness: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

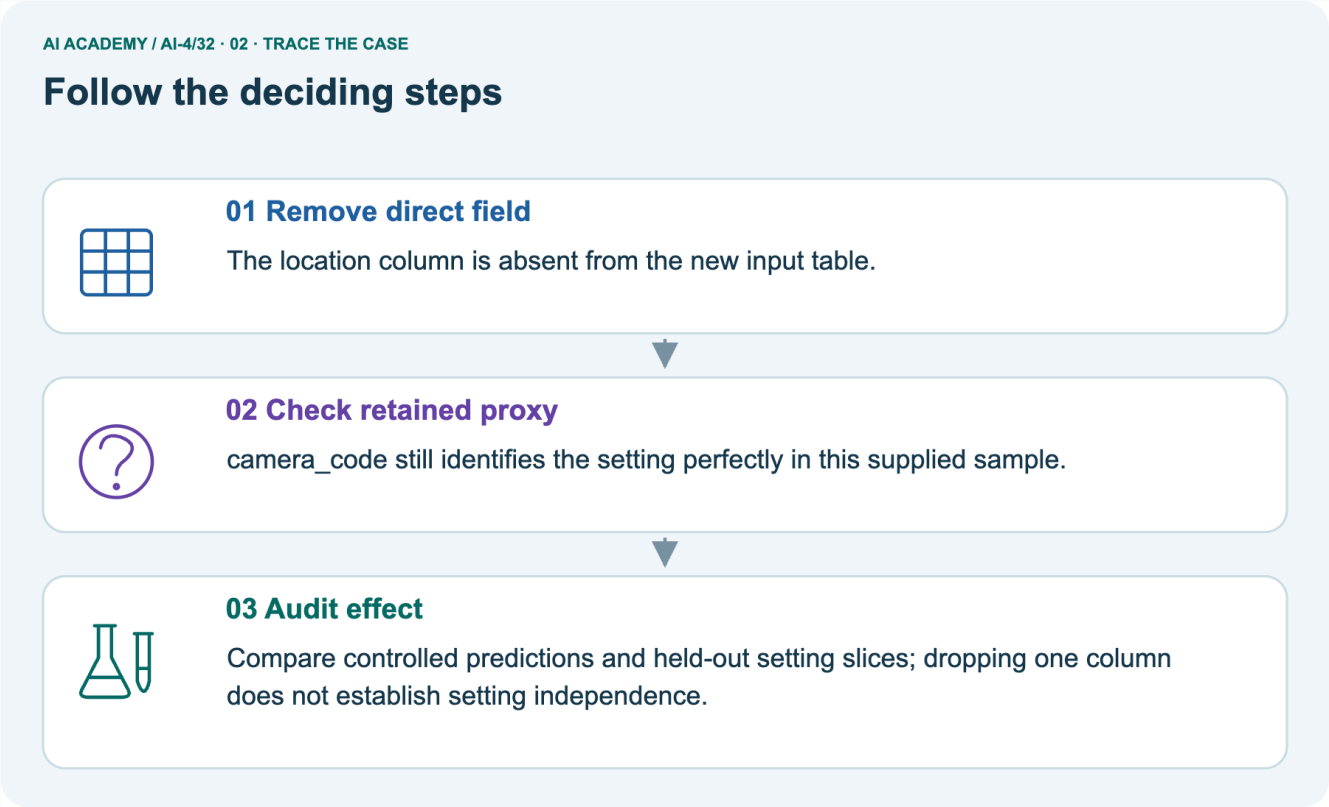
Removing an obvious setting field may leave another field that carries the same information. Supplied facts: Toy table has location=greenhouse/desk and camera_code=G/D. Every G record is greenhouse; every D record is desk.

The team drops location. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



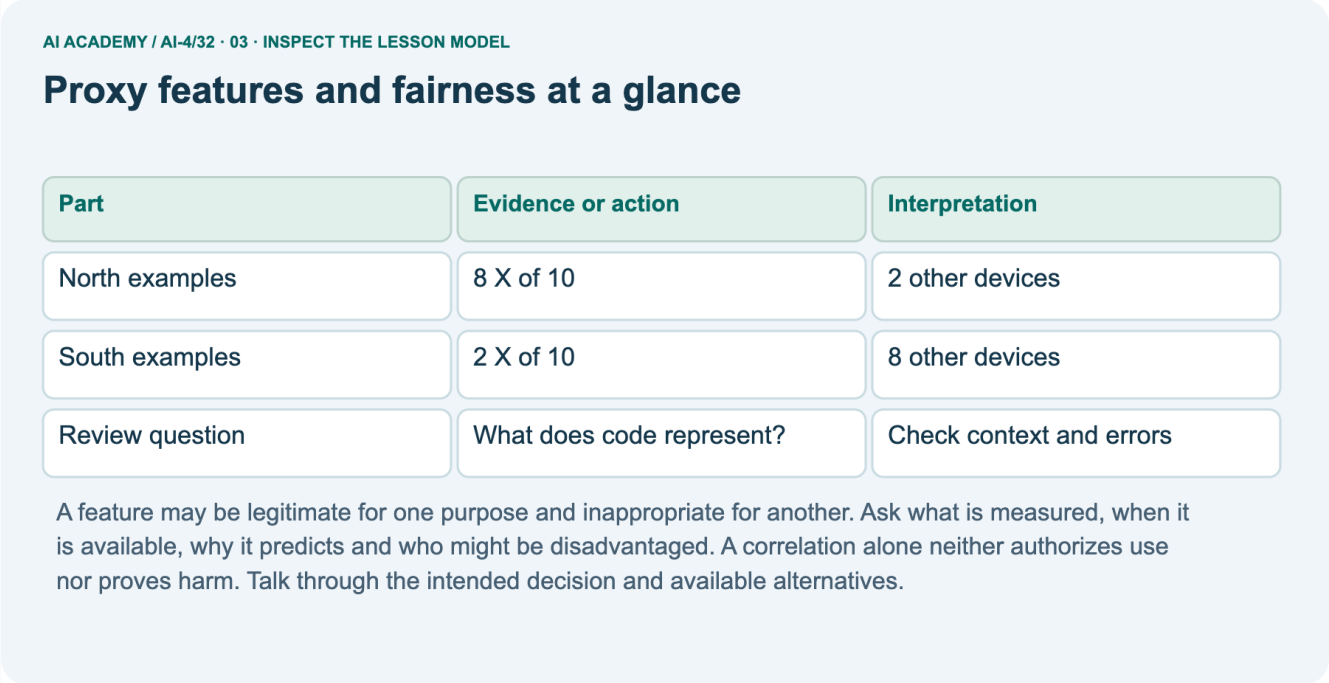
Follow the labelled picture with the matching explanation. Open the vector diagram

Remove direct field: The location column is absent from the new **input** table. → Check retained proxy: camera_code still identifies the setting perfectly in this supplied sample. → Audit effect: Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Proxy features and fairness at a glance



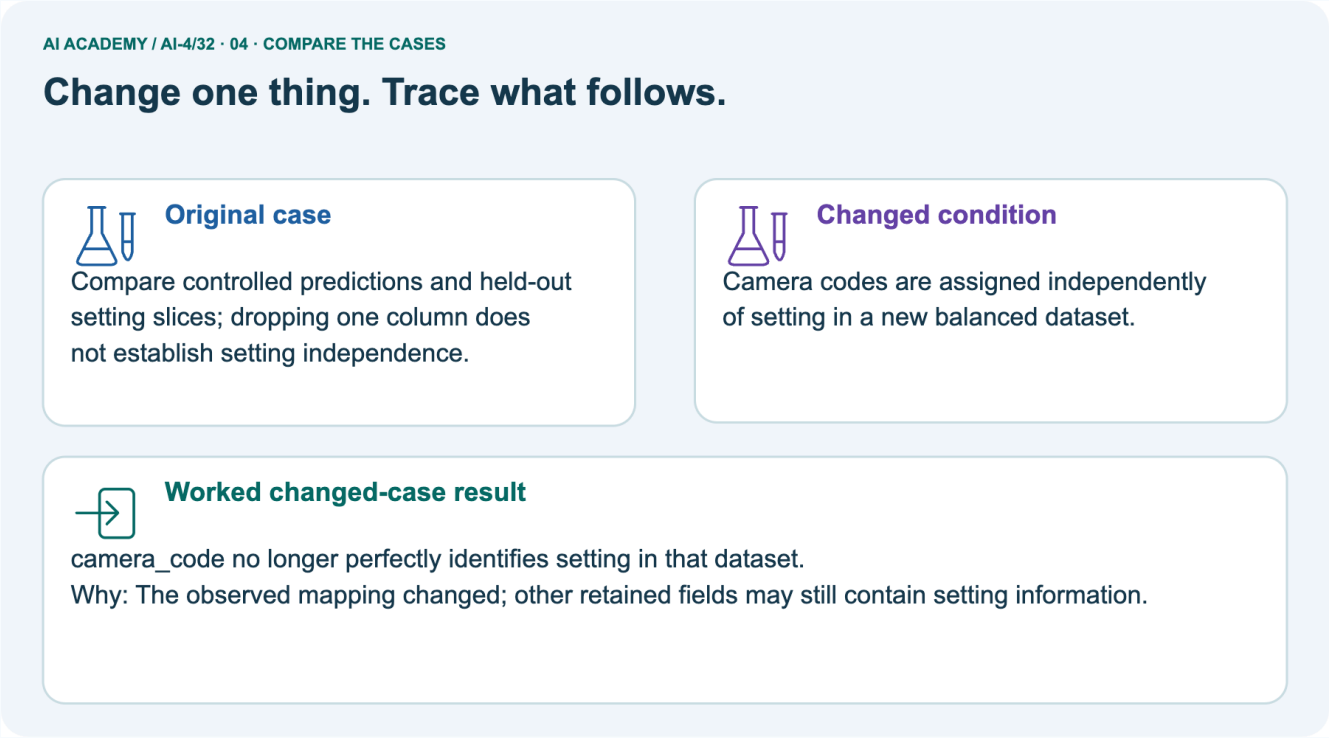
Follow the labelled picture with the matching explanation. Open the vector diagram

Proxy features and **fairness** at a glance. North examples | 8 X of 10 | 2 other devices. South examples | 2 X of 10 | 8 other devices. Review question | What does code represent? | Check context and errors

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

Changed condition: Camera codes are assigned independently of setting in a new balanced dataset. Expected result: camera_code no longer perfectly identifies setting in that dataset.

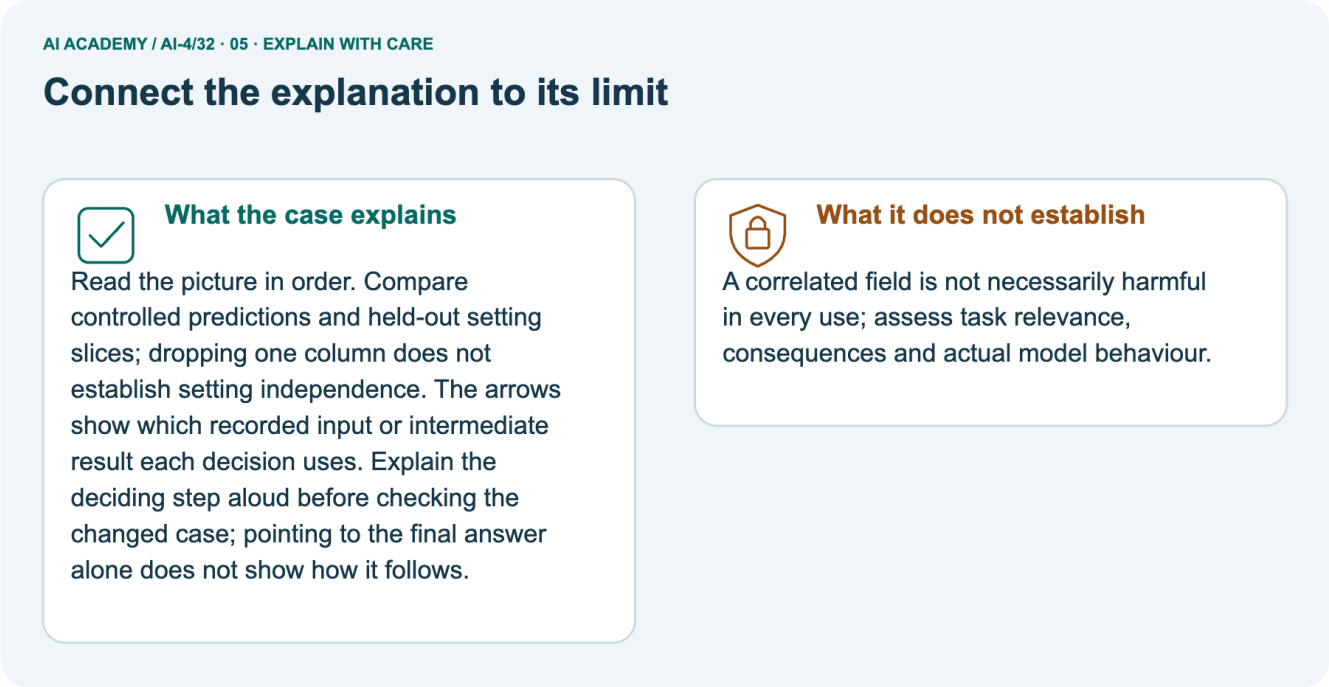
Reason: The observed mapping changed; other retained fields may still contain setting information.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The observed mapping changed; other retained fields may still contain setting information.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

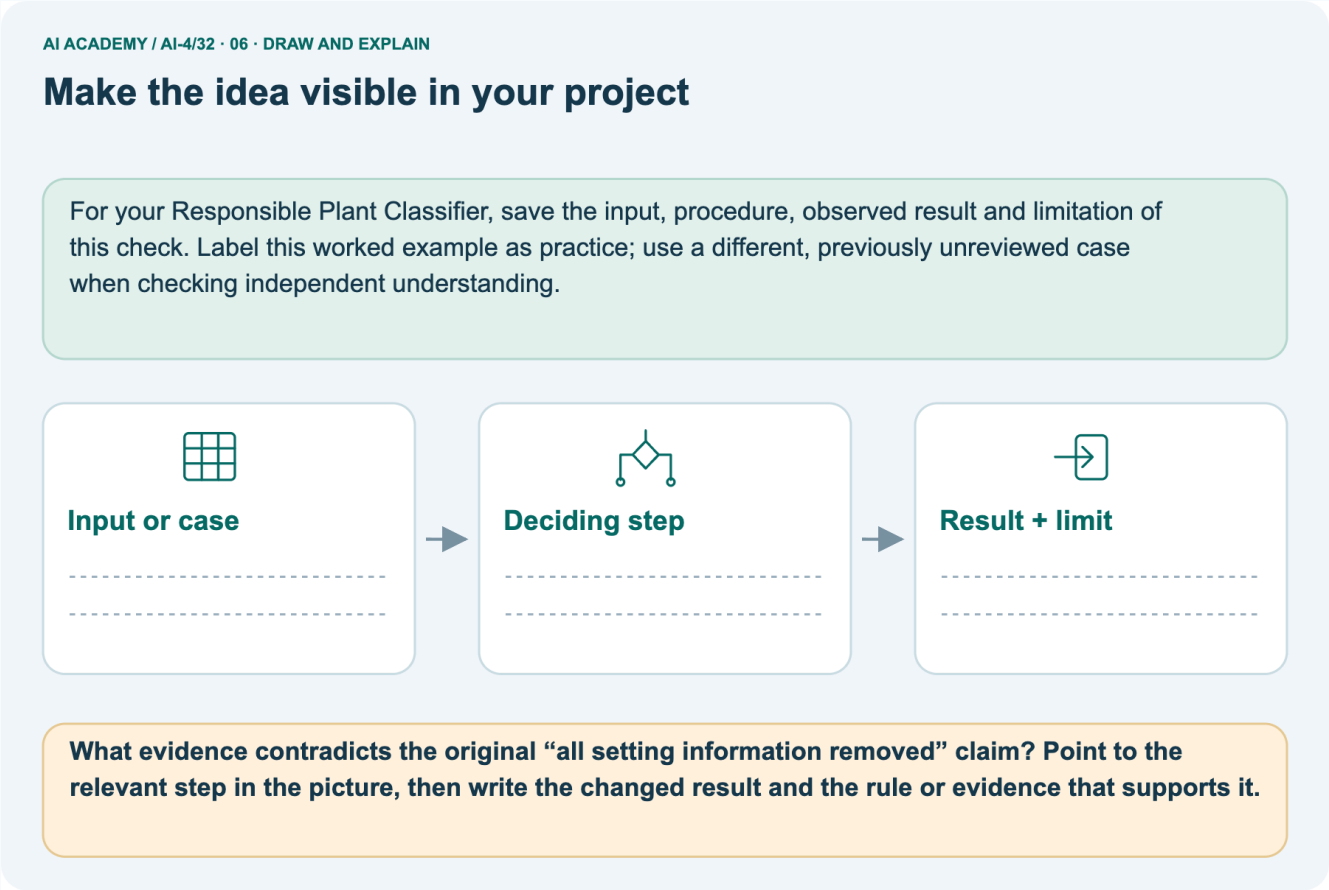
Limit: A correlated field is not necessarily harmful in every use; assess task relevance, consequences and actual model behaviour.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A correlated field is not necessarily harmful in every use; assess task relevance, consequences and actual model behaviour.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What **evidence** contradicts the original “all setting information removed” claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The retained G/D code maps exactly to greenhouse/desk, so the supplied table still exposes setting information. Changed-case check: camera_code no longer perfectly identifies setting in that dataset.

The observed mapping changed; other retained fields may still contain setting information.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Proxy Features and Fairness: the case

02. Follow the deciding steps

03. Proxy features and fairness at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.